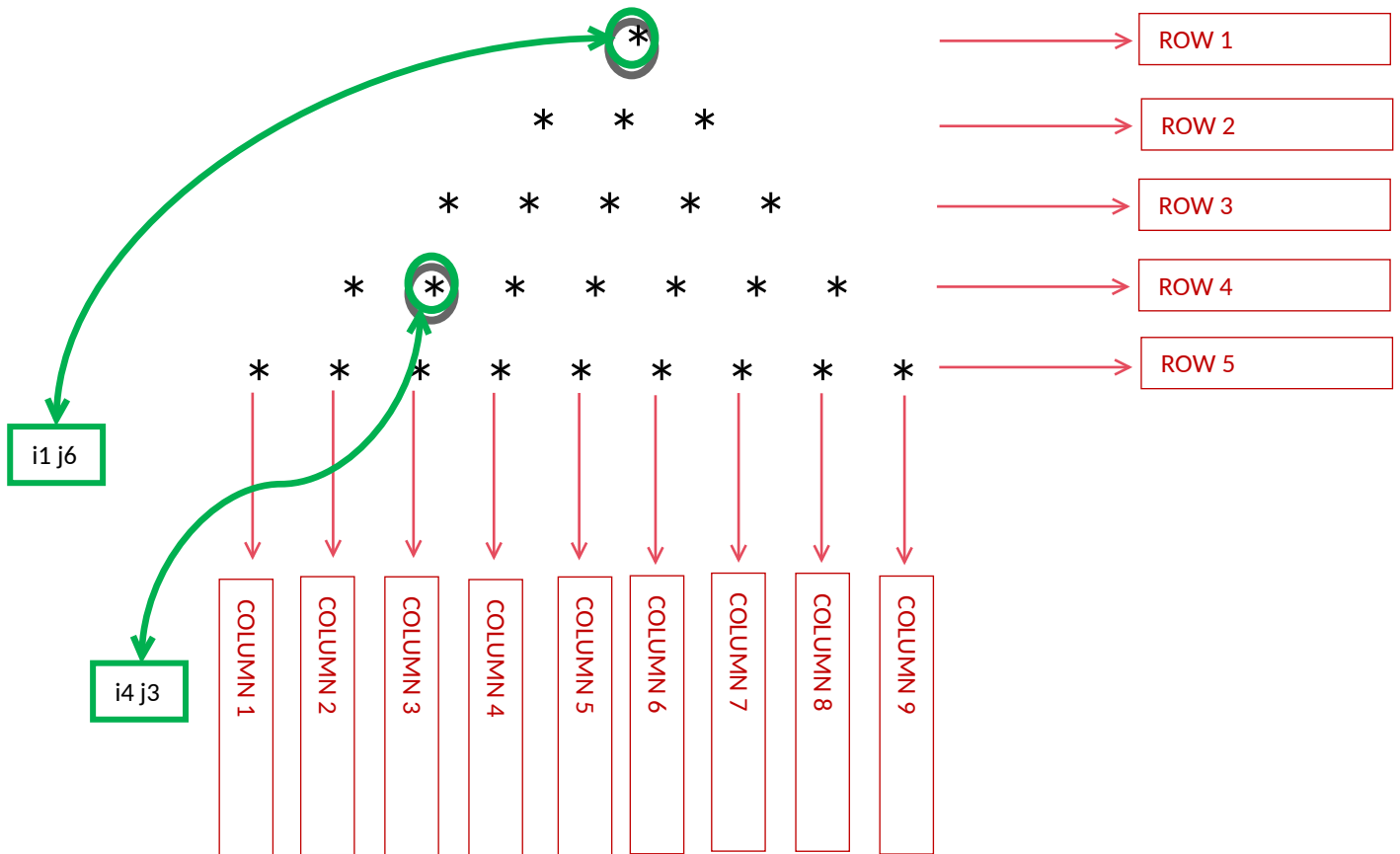


WEEK 01 MODULE
Programming Practice

TOPIC : 1 'Star Pattern'

1. PYRAMID



1. Problem Analysis :-

i. Pattern Structure:

- The output forms a centered pyramid.
- Unlike the Triangle and Inverted Triangle patterns, the stars are not aligned to the left side.
- spaces are printed before the stars so that the pattern appears centered.

ii. Execution Flow:

Step 1: Observe the Grid Concept:

- For $n = 5$

```

      *
     * * *
    * * * * *
   * * * * * *
  * * * * * * *
 * * * * * * *

```

The pattern grows symmetrically from the center.

As we move downward:

- The number of spaces decreases.
- The number of stars increases.

Both of these changes occur simultaneously.

Therefore, this pattern requires two different calculations in every row.

- Each cell in the grid represents a unique coordinate (i, j), where i denotes the row index and j denotes the column index.

-The pattern can be viewed as two regions:

✦ **Region 1: Leading Spaces**

These spaces shift the stars toward the center.

✦ **Region 2: Stars**

These form the visible pyramid.

For n = 5

Row 0 : [][*]

Row 1 : [][* * *]

Row 2 : [][* * * * *]

Row 3 : [][* * * * * * *]

Row 4 : [][* * * * * * * * *]

As the row index increases:

- Spaces decrease by 1.
- Stars increase by 2.

-> Row Analysis :-

Row Index (I)	0	1	2	3	4
Spaces	4	3	2	1	0
Stars	1	3	5	7	9

The pyramid is controlled by two formulas:

$$\text{Spaces} = n - i - 1$$

$$\text{Stars} = 2 \times i + 1$$

Step 2: Traverse the grid row by row.

- The outer loop is responsible for generating rows.

Step 3: For every row, calculate the number of leading spaces required.

- The number of spaces depends on the current row index.
- The relationship is:

$$\text{Spaces} = n - i - 1$$

Step 4: Traverse the required number of positions and print spaces.

- This shifts the stars toward the center.
- The number of spaces decreases as the row index increases.

Step 5: After printing spaces, calculate the number of stars required.

- The relationship is:

$$\text{Stars} = 2 \times i + 1$$

Step 6: Traverse the required number of positions and print stars.

- The number of stars increases by 2 in every row.

Step 7: After all stars of the current row have been printed, move the cursor to the next line.

- This ensures that the next row starts on a new line.

Step 8: Repeat the process for all rows until the outer loop terminates.

2. Condition Analysis

No conditional statements are required.

There is no need for if(...)

- Because every position visited by the space loop prints a space.
- Every position visited by the star loop prints a star.
- The pattern is controlled entirely through loop limits.

3. Core Logic Transformation

Pyramid Pattern:-

```
for(int j = 0; j < 2 * i + 1; j++)
```

- Meaning:

Stars increase by 2 in every row.

Additionally :-

```
for(int j = 0; j < n - i - 1; j++)
```

- controls the alignment of the pattern.

4. Program :-

```
package week1_module;
import java.util.Scanner;
public class StarPyramid {

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter the number of rows: ");
        int n = sc.nextInt();

        for(int i = 0; i < n; i++) {

            // Print spaces
            for(int j = 0; j < n - i - 1; j++) {
                System.out.print(" ");
            }

            // Print stars
            for(int j = 0; j < 2 * i + 1; j++) {
                System.out.print("* ");
            }

            System.out.println();
        }

        sc.close();
    }
}
```

5. Program Output :-

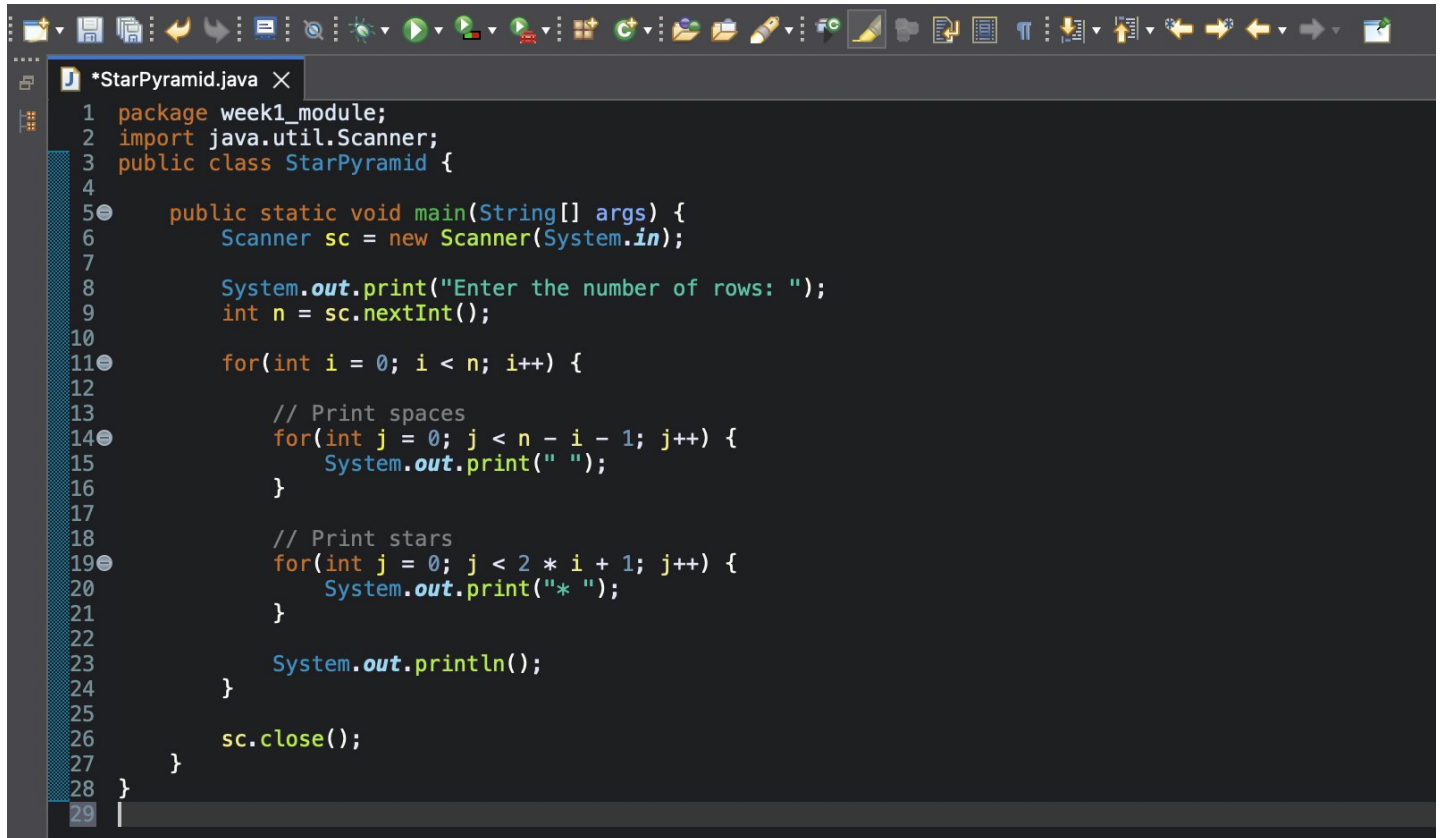
Enter the number of rows: 5

```

  *
 * *
* * *
* * * *
* * * * *
* * * * * *
```

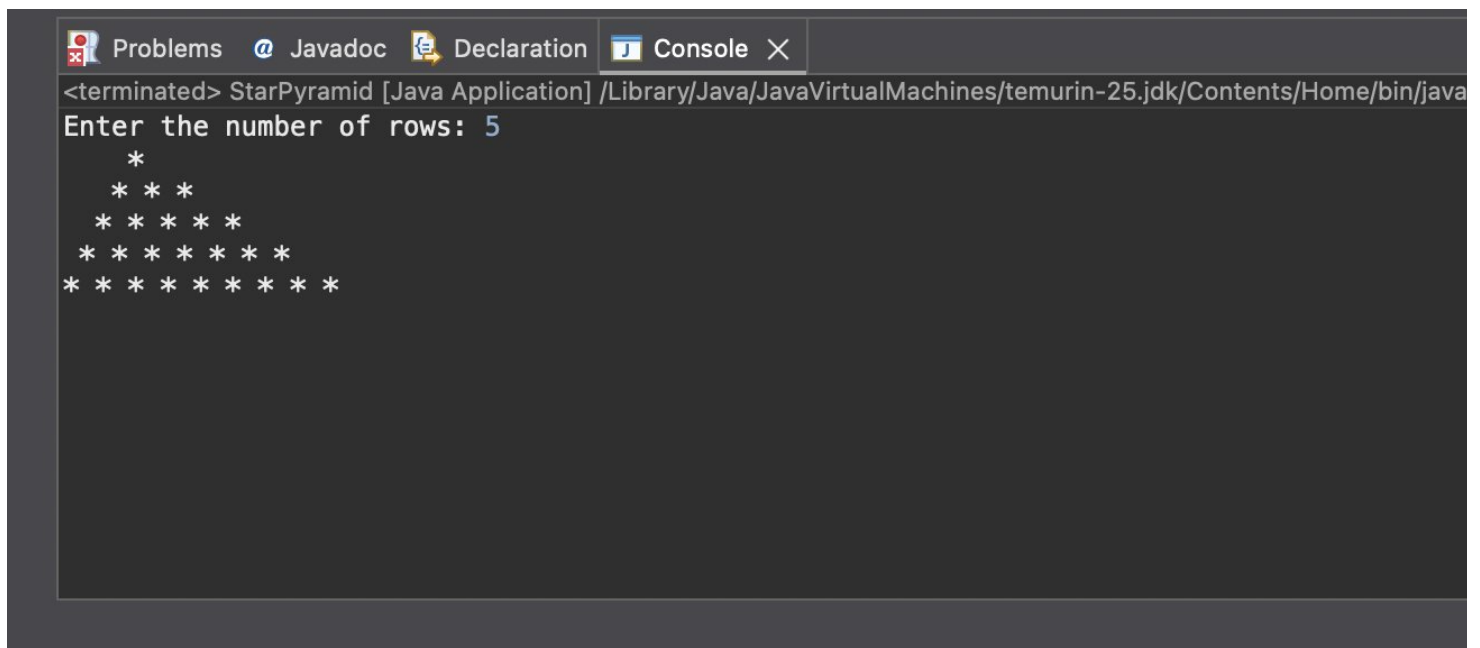
6. ScreenShot :-

i. Program :-



```
1 package week1_module;
2 import java.util.Scanner;
3 public class StarPyramid {
4
5     public static void main(String[] args) {
6         Scanner sc = new Scanner(System.in);
7
8         System.out.print("Enter the number of rows: ");
9         int n = sc.nextInt();
10
11         for(int i = 0; i < n; i++) {
12
13             // Print spaces
14             for(int j = 0; j < n - i - 1; j++) {
15                 System.out.print(" ");
16             }
17
18             // Print stars
19             for(int j = 0; j < 2 * i + 1; j++) {
20                 System.out.print("* ");
21             }
22
23             System.out.println();
24         }
25
26         sc.close();
27     }
28 }
29
```

ii. Program's Output :-



```
<terminated> StarPyramid [Java Application] /Library/Java/JavaVirtualMachines/temurin-25.jdk/Contents/Home/bin/java
Enter the number of rows: 5
    *
   * * *
  * * * * *
 * * * * * * *
* * * * * * * * *
```